Chapter 1
Introduction



Statistics and metrology, that is, the science of measurement, permeate mod-ern engineering, society, and culture. This chapter provides a brief historical perspective on the importance of metrology, along with introducing the core concept of *uncertainty*: a key element that spans both statistics and metrology. The importance of uncertainty is demonstrated through two contemporary case studies. First, this chapter considers how uncertainty may have adversely affected the outcome of pressure measurements on footballs in a 2015 sports scandal often referred to as "Deflategate." Second, the importance of an accurate measurement model in obtaining reliable estimates of case fatality rate during a pandemic is discussed.

1.1 Measurement Uncertainty: Why Do We Care?

Measurements are an important part of everyday life. Measurements drive decision-making in nearly all aspects of modern society. Do you ever question the validity of a measurement result? You should, considering that every measurement has uncertainty. Knowledge of this associated uncertainty is *necessary* for making *informed* decisions. Yet, uncertainty analysis and propagation constitute a subject area that rarely receives proper attention in science and engineering curriculums or in public discourse.

Uncertainty analysis and propagation are central to metrology, the science of measurement. And metrology is fundamentally interlinked with statistics. This textbook helps bridge the gap between statistics, metrology, and uncertainty analysis, not just for the measurement practitioner, but also for those who utilize measurement data. In this chapter, a brief historical backdrop is provided, along with two introductory case studies based on contemporary issues. Case studies demonstrate the importance of a holistic approach to measurement uncertainty and highlight several concepts and methods that will be introduced in later chapters.

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Fig. 1.1 A replica of the ancient Egyptian cubit rod. A cubit is a unit of length equivalent to the distance between the elbow and the tip of the middle finger of the ruling Pharaoh at the time

1.2 The History of Measurement

The importance of proper measurement and its associated measurement uncertainty have long been recognized. The ancient Egyptians and ancient Mesopotamians established the earliest known recorded system of measure while constructing the pyramids and other architectural feats in the 4th to 3rd millennia BCE (Clarke and Engelbach 1990). Figure 1.1 shows an example of a cubit rod used by the ancient Egyptians. The cubit rod was used as a unit of length and represented the length of a Pharaoh's forearm. In ancient Israel, a well-established system of measurement was used to ensure that trade of food items and other goods was carried out fairly and justly. Prior to the introduction of a standardized system, some would intentionally offset weights and measures to try to cheat their neighbors in the sale or trade of goods.

Prior to 500 BCE, the ancient Greeks developed a system of official weights and measures and employed a method of calibration using reference standards. The Roman Empire adapted the earlier Hellenic system to create a well-documented, sophisticated system of measurement that employed standards and calibration in commerce, trade, and engineering (Smith 1851). However, many of the official weights and measures varied from region to region. In the third century BCE, the Greek librarian and scientist Eratosthenes measured the circumference of the Earth simply by observing the lengths of shadows in two locations. He determined the circumference to be 250,000 stades. The stade was a unit based on the length of a typical Greek stadium, but there were several regional variations on the definition of a stade (Walkup 2005). Unfortunately, because of the different definitions in use at the time, it may never be known exactly how close Eratosthenes's measurement came to the modern accepted circumference of the Earth. Not until many centuries later did weights and measures become internationally standardized.

1.3 Measurement Science and Technological Development

Throughout history, advances in measurement science and standards have been a prerequisite for the practical implementation of scientific developments. For example, the concept of interchangeable parts, which revolutionized manufacturing and led to numerous other technological advances, could only be implemented successfully with improvements in measurement science. In the late 1700s and early 1800s, complex mechanical devices, such as firearms, required careful hand-

fitment of parts by experienced gunsmiths. In 1801, when Eli Whitney presented his concept of interchangeable parts for a musket to the United States War Department (Hays 1959), a key framework, such as dimensional standards, was not yet available. Unbeknownst to the War Department, the muskets presented were specially prepared and individual parts were not standardized or interchangeable with other muskets of the same type. It took decades to successfully realize the vision of interchangeable parts for modern manufacturing. In fact, gauge blocks, a type of dimensional standard introduced in the 1890s by Swedish machinist C.E. Johannson (Althin 1948), were a key facilitator for modern production and fabrication methods, enabling the use of interchangeable parts and common tooling for assembly line manufacturing.

The importance of measurement standardization and measurement traceability for technological advancement and economic growth was realized early in the Technological Revolution. In 1875, 17 countries, including the USA, signed the Metre Convention, which established the International Bureau of Weights and Measures (BIPM) and defined international standards for mass and length.

Improvements in standards and measurement would continue to drive major revolutions in machining, fabrication, and production of industrial, commercial, and consumer goods throughout the 1900s. The advent of digital computing in the late 1940s, along with advances in air and space travel throughout the last century, necessitated improvements in all areas of measurement, with associated reductions in uncertainty. Despite the agreed-upon importance of a universal system of units and maintenance of consistent standards, treatment of measurement data and its associated uncertainties remained relatively ambiguous until the 1990s. To this day, measurement uncertainty is often misunderstood by engineers and scientists performing measurements.

The criticality of accurate measurements in the marketplace has never been greater. Measurement inaccuracies in food and fuel purchases alone place billions of consumer dollars at risk each year. Measurement uncertainties in manufacturing increase the risk of accepting bad product or rejecting good product, each resulting in lost productivity and profits. Measurement uncertainties in medical diagnostics can result in both missed or incorrect diagnoses, with major public health implications. Measurements are the basis for legal decisions and evidence in trials and form the basis for science-based policies across the globe. The accuracy of such measurements relies heavily on measurement standardization and measurement traceability and is the forefront of topics discussed in this handbook. We will demonstrate the importance of measurements, and their associated uncertainties, with two modern real-world case studies highlighting important measurement uncertainty concepts detailed in later chapters.

1.4 Allegations of Deflated Footballs (“Deflategate”)

During the 2014 American Football Conference (AFC) Championship Game on January 18, 2015, between the Indianapolis Colts and the New England Patriots of the National Football League (NFL), allegations arose regarding the New England Patriots intentionally deflating their game balls to provide an unfair competitive

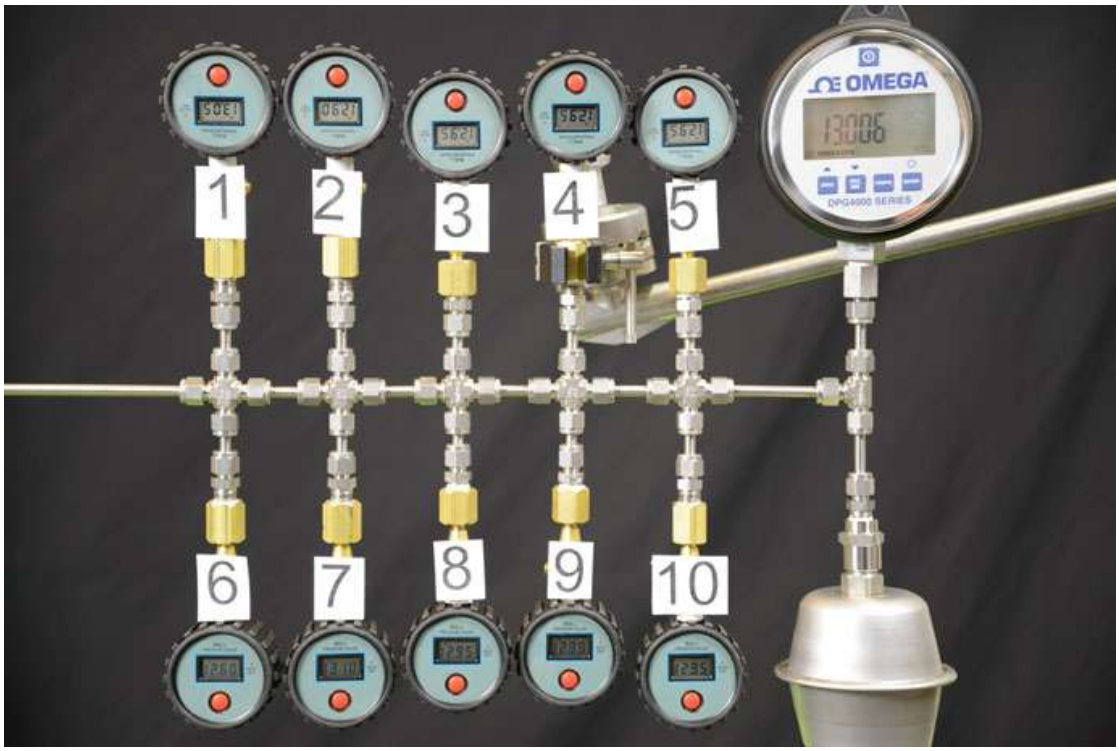


Fig. 1.5 "Calibrated" master gauge experiment. Traceability was based on calibration provided from an unaccredited laboratory (Exponent 2015)

In addition, the so-called calibration of the handheld game-day pressure gauges after the fact is not a valid practice (see Chap. 3) and only constitutes a characterization. There is no way to guarantee that the gauges performed the same on game-day due to drift and other factors. The uncertainty of the "master" gauge, and uncertainties in general, was not considered or incorporated. An uncertainty or tolerance must be assigned to a unit under test (UUT) during a valid calibration.

While other evidence, such as interviews with players, officials, and equipment personnel, along with text message conversations ultimately weighed on the outcome of the investigation and sanctions by the NFL, the centerpiece of the case was untraceable measurements of internal football air pressure using equipment with an unacceptably low test uncertainty ratio. As seen in this example, concepts of measurement uncertainty, uncertainty propagation, calibration, and traceability have important implications in sports and legal investigations but are unfortunately not always applied properly. Decisions made on measurement data are only as good as the uncertainties that come with it.

1.5 Fatality Rates During a Pandemic

An infectious disease is spreading around the globe, with dire predictions of lethality. Shortly after the World Health Organization (WHO) announces a Phase 6 pandemic alert and the USA declares a Public Health Emergency, fatality

establishments have been ordered closed. In a 4-week period, an unprecedented 22 million people have filed for unemployment. Extreme measures have been implemented in nearly every state, with fines or imprisonment possible for those violating stay-at-home orders. Such measures have resulted, at least in part, from the WHO's initial estimated fatality rate and similar estimates that predicted 2.2 million deaths in the USA if no further actions were taken (Ferguson et al. 2020). Economists at the International Monetary Fund have predicted these measures will result in the worst economic downturn since the Great Depression (Gopinath 2020).

The infectious disease is coronavirus disease 2019 (COVID-19), caused by the novel coronavirus severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). The WHO's initial estimate came from applying measurement "Model 1" in Eq. (1.4) to data coming from China. Some, including the director of the U.S. National Institute of Allergy and Infectious Diseases (NIAID), recognized early on that such estimates could significantly overestimate the actual case fatality rate (Fauci et al. 2020, italicization added):

> If one assumes that the number of asymptomatic or minimally symptomatic cases is several times as high as the number of reported cases, *the case fatality rate may be considerably less than 1%*. This suggests that the overall clinical consequences of Covid-19 may ultimately be more akin to those of a severe seasonal influenza (which has a case fatality rate of approximately 0.1%) or a pandemic influenza (similar to those in 1957 and 1968) rather than a disease similar to SARS or MERS, which have had case fatality rates of 9 to 10% and 36%, respectively.

Indeed, revised models and studies report much lower fatality rates of 0.657% (range: 0.389–1.33%) when undiagnosed cases are estimated and then included in the estimate of fatality rate (Verity et al. 2020). Note that Verity et al. (2020) refer to this fatality rate as the "infection fatality ratio" to differentiate the denominator from positive confirmed cases, which, has become accepted terminology but is essentially equivalent to the CFR as defined by Dicker et al. (2012) and Fauci et al. (2020). Preliminary, non-peer reviewed serological studies, looking for the presence of antibodies due to immune response to SARS-CoV-2 in previously undiagnosed cases (the bottom two regions in Fig. 1.6), have already suggested substantially lower case fatality rates. A serological survey of an entire municipality in Germany estimated a case fatality rate of 0.37% (Regalado 2020), and a serological study in Santa Clara County, California estimated a fatality rate of 0.12–0.2% (Bendavid et al. 2020; Mallapaty 2020). More reliable data and improved measurement models will ultimately provide more realistic estimates of the true COVID-19 fatality rate.

1.6 Summary

Ancient societies realized measurement improvement and standardization were necessary for stability and development. To this day, measurement advances are closely tied to new technology implementation. And yet, measurements and their associated uncertainties are poorly understood by the general public and receive only limited consideration in college and university programs.

An understanding of measurement uncertainty is critically important in all fields of science, engineering, statistics, healthcare, economics, business, and any discipline that relies on measurement or measurement data for decision-making. Measurement uncertainty also affects our everyday lives. Failure to properly account for measurement uncertainty has led to major engineering disasters, loss of human life, and trillions of dollars in economic losses.

As we have seen in two contemporary case studies, measurement uncertainty has important implications in sports, legal settings, and the study of diseases. These case studies have presented, at a high level, a number of concepts and methods, blending statistics and metrology, that are the focus of this book.

1.7 Related Reading

The history of metrology and its role in modernizing the world is an interesting and ongoing area of study that cannot be given fitting coverage in this book. *The Science of Measurement: A Historical Survey* by Klein (1988), first published in 1974, provides a detailed look into the history of unit standardization. With a focus on historical developments in the International System of Units (SI), Klein's writing style and storytelling through the eyes of historical figures provide for a compelling read. Understandably, there are gaps in more recent developments in metrology given it was first written in 1974. Klein also offers historical perspectives on measurement developments in the areas of thermal and ionizing radiation, which are not found in other texts.

Lugli (2019) describes how revolutions in measurement processes in medieval Europe brought about revolutions in other areas of science and society. With a focus on medieval Italy up through the Renaissance, Lugli paints an extraordinary picture of the impacts of measurement standardization on modernization from a sociological perspective. Lugli provides numerous rich primary sources that help illustrate the tumultuous development of measurement standardization in medieval Italy.

For a reference text on historical and modern units of measurement, Gyllenbok (2018) offers perhaps the most comprehensive collection to date of measurement systems and units, by country, from ancient to modern times. Consisting of three volumes, the encyclopedia is well-organized and would serve as an excellent desk reference for the modern metrologist or historian. Gyllenbok also provides an immense collection of references and related reading.

Fluke Calibration also offers a short (~40 min) free webinar, titled "A Brief History of Metrology and Advancements in the Science of Measurement," available at https://us.flukecal.com/history-of-Metrology.

Another recommended reading that provides a brief historical perspective on metrology with a detailed bibliography and serves as a useful companion to this entire textbook is *The Handbook of Metrology* by Bucher (2012). While the primary audience is metrologists and calibration laboratory personnel, Bucher's coverage of quality systems and metrology concepts is useful for all practitioners of metrology.

As seen in the second case study of this chapter, metrology, statistics, and uncertainty are critically important in the field of epidemiology. While this textbook provides the fundamental tools for performing uncertainty analyses using any measurement model, epidemiological models and studies are not a focus of this book. For further reading in this area, *Modern Epidemiology* by Rothman et al. (2008) provides thorough coverage of statistical concepts for application in epidemiology, along with discussions on means of improving the accuracy of studies in this field. Stewart (2016) provides an excellent introduction to statistics concepts for those with no prior experience and basic concepts for applying statistics to epidemiological studies.

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